

COURSE TITLE : ANALOG CIRCUITS
COURSE CODE : 3211
COURSE CATEGORY : B
PERIODS/WEEK : 5
PERIODS/SEMESTER: 75
CREDITS : 5

TIME SCHEDULE

Module	Topics	Periods
1	Amplifiers	20
2	Oscillators & Multivibrators	19
3	Wave shaping circuits	16
4	Operational amplifier	20
TOTAL		75

COURSE OUTCOME

Module	G.O.	On completion of the study of this module the student will be able
1	1	To understand the concept of transistor amplifier
	2	To understand the concept of feedback amplifier
2	1	To know the principle of oscillator circuits.
	2	To understand the operation of Multivibrators
3	1	To understand the linear wave shaping circuits
	2	To know the non-linear wave shaping circuits
4	1	To understand Basics of OP amp circuits
	2	To know the applications of OP amps

On completion of the study the student will be able

MODULE I AMPLIFIERS

1.1.0 To understand the concept of transistor amplifier.

- 1.1.1 To study the behavior of the CE amplifier with potential divider biasing.
- 1.1.2 To determine the DC & AC load line of CE amplifier.
- 1.1.3 To calculate the voltage gain, current gain, power gain and I/O impedance.
- 1.1.4 To study the working of multistage amplifiers.
- 1.1.5 To study the different methods of inter-stage coupling.
- 1.1.6 To study the working Principles and features of RC coupled, Transformer coupled and direct coupled multistage transistor amplifier.
- 1.1.7 To compare the performance of RC coupled, transformer coupled and direct coupled amplifiers.
- 1.1.8 To study “frequency response” and “bandwidth” of RC Coupled amplifiers.
- 1.1.9 To study the characteristics and applications of Emitter Follower.

1.2.0 To understand the concept of feedback amplifier.

- 1.2.1 To describe the positive and negative feedback
- 1.2.2 To derive the expression for the gain of feedback amplifier.

MODULE II OSCILLATORS & MULTIVIBRATORS

2.1.0 To understand the principle of oscillator circuits.

- 2.1.1 To state the Barkhausen criterion for oscillation.
- 2.1.2 To describe the working of RC oscillators, RC phase shift oscillator and Wein bridge oscillator.
- 2.1.3 To list the applications of RC oscillators.
- 2.1.4 To explain the basic principle of LC oscillators.
- 2.1.5 To explain the working of Hartley and Colpitt's oscillators.
- 2.1.6 To list the applications of LC oscillators.
- 2.1.7 To explain the operation of crystal oscillator.
- 2.1.8 To list the advantages and applications of crystal oscillator.

2.2.0 To understand the operation of Multivibrators

- 2.2.1 Name the types of multivibrators.
- 2.2.2 To describe the operation of Astable multivibrator using transistors.
- 2.2.3 To derive the expression for the frequency of oscillation.
- 2.2.4 To explain the operation of monostable multivibrator using transistors.
- 2.2.5 To describe the operation of Bistable multivibrator using transistors.
- 2.2.6 To understand the timer IC 555
- 2.2.7 To list the applications of multivibrators.

MODULE III WAVE SHAPING CIRCUITS

3.1.0 To understand the linear wave shaping circuits

- 3.1.1 To know the different wave shapes.
- 3.1.2 To define the characteristics of pulse wave form - rise time, fall time and tilt.
- 3.1.3 To study RC differentiating and Integrating circuits.

- 3.1.4 To state the conditions for proper integration and differentiation.
- 3.1.5 List the applications of integrator and differentiator circuits.

3.2.0 To understand the non-linear wave shaping circuits

- 3.3.1 To classify the diode clipping circuits.
- 3.3.2 To explain the operation of series shunt and biased type clipping circuits with wave forms.
- 3.3.3 To explain the operation of Zener diode clipper circuits.
- 3.3.4 To define the term clamping.
- 3.3.5 To explain the operation of various types of diode clamping circuits.
- 3.3.6 List the applications of clamping circuits.

MODULE IV OPERATIONAL AMPLIFIER

4.1.0 To understand Basics of OP amp circuits

- 4.1.1 To illustrate internal block schematic of Op amp
- 4.1.2 To explain different types of OP amp parameters
- 4.1.3 To describe the characteristics of OP amp

4.2.0 To understand the Op-amp applications

- 4.2.1 To illustrate inverting and non-inverting amplifier
- 4.2.2 To explain voltage follower
- 4.2.3 To describe adder and subtractor
- 4.2.4 To describe integrator and differentiator
- 4.2.5 To explain Current to Voltage converter
- 4.2.6 To explain Voltage to Current converter
- 4.2.7 To describe Precision rectifier
- 4.2.8 To describe comparators
- 4.2.9 To illustrate Schmitt trigger
- 4.2.10 To describe error detector
- 4.2.11 To explain instrumentation amplifiers

CONTENT DETAILS

MODULE I

Action of transistor as an amplifier – single stage C.E amplifier circuit with potential divider biasing – AC and DC load line – voltage gain, current gain, power gain and I/O impedance - Multistage Amplifiers - inter-stage coupling methods- RC coupled, Transformer coupled and direct coupled multistage transistor amplifier- frequency response and bandwidth of RC Coupled amplifiers- Emitter Follower- feedback amplifier- positive and negative feedback- the gain of feedback amplifier

MODULE II

Oscillators-Classification– Barkhausen criteria–RC Oscillators: RC phase shift oscillator- Wein bridge oscillator – LC oscillators- Hartley oscillator- Colpitt's Oscillator-applications of RC and LC oscillators – crystal oscillator – advantages and applications - Types of multivibrators – Astable Multivibrator - expression for frequency of Oscillation- Monostable multivibrator - factors affecting pulse width- Bistable multivibrator - working with base and collector wave forms-timer IC 555-Applications of multivibrators

MODULE III

Types of wave forms – ideal pulse wave form – rise time- fall time- and tilt – RC differentiator - RC integrator –Applications-Clipping circuits - diode clippers – series-shunt- biased and combinational clipper circuits - Zener diode clipper circuits - Clamping circuits - diode clammers – positive- negative- and biased type with sine and square wave input

MODULE IV

Operational Amplifiers – internal block schematic of Op amp - Op amp parameters-characteristics of OP Amp -inverting and non-inverting amplifier -voltage follower- integrator- differentiator- adder-subtractor –current to voltage - voltage to current converter– comparators - Precision rectifier- Schmitt trigger- error detector - instrumentation amplifiers

REFERENCES

1. J.B.Gupta,Electronic circuits and devices, S.K.Kataria and sons
2. Roy Choudhary,Linear integrated circuits,New Age publications
3. Millman and Halkias,Electronic devices and circuits,TataMcGraw-Hill
4. Ramakant Gayakward , Opamps and Linear Integrated Circuits , Pearson education
5. Floyd , Electronic devices , Pearson education